

SAMUEL JOHNSTON

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Software Developer specialized in transforming complex Remote Sensing algorithms into fast, production-ready pipelines. I build robust and scalable processing solutions, which accelerate ML and Data Science workflows.

SKILLS

Languages	Python, C, C++, Javascript	Deep Learning	PyTorch, Scikit-Learn
Data	Xarray, Zarr, Dask	Natural Languages	English, French
Geospatial	Rasterio, GEE, QGIS, STAC		

PROFESSIONAL EXPERIENCE

H2O Geomatics <i>ML & Data Developer</i>	Kitchener, ON <i>September 2024 - present</i>
· Operational EO Production : Orchestrated a 3D-CNN pipeline in HPC environment to cloud-gap-fill a 20+ year daily dataset for 1,391 global lakes, delivering high-quality data to stakeholders.	
· Data Pipeline Optimization : Refactored a 1.5 PB of MODIS reflectance pipeline; achieved a 12x speedup and 16x memory reduction .	
<i>Machine Learning Researcher</i>	<i>Jan 2023 - Sept 2023</i>
· ML Ops & Reproducibility : Accelerated model training cycles by an estimated 60%, by leveraging WandB to configure hyperparameter sweeps and for experiment tracking and versioning.	
ECMWF <i>Visiting Scientist</i>	Reading, UK <i>Oct 2023 - Dec 2023</i>
· Numerical Computation : Conducted high-resolution Numerical Weather Prediction (NWP) experiments on massive HPC clusters to optimize lake model parameterization.	
Lakes Environmental Software <i>Data Developer</i>	Waterloo, ON <i>Sept 2022 - Dec 2022</i>
· Data Pipeline Development : Built a production-ready ETL pipeline to ingest and harmonize Sentinel-1 SAR and Dynamic World data, producing 2,000+ geo-referenced training patches.	
<i>Machine Learning Researcher</i>	<i>Sept 2021 - April 2022</i>
· Machine Learning Research : Research on Deep Learning for Computational Fluid Dynamics (CFD) simulations. (Pub: Appl. Sci. 2023).	

EDUCATION

University of Waterloo <i>Bachelors of Computer Science (Honours), Diploma in Sustainability</i> <i>Research Assistant</i>	Waterloo, ON <i>Sept 2019 - April 2024</i> <i>May 2024 - Aug 2024</i>
· Developed Markov Model to analyze agricultural trends within the Annual Crop Inventory dataset.	
<i>Research Assistant</i>	<i>Jan 2021 - Aug 2021</i>
· Developed Random Forest model, predicting nutrient loading, achieving R^2 0.6–0.8. (Pub: Earth's Future 2023).	

PROJECTS

Automated Boulder and Cliff Detection Platform (In Development)
Platform to detect potential boulders and cliffs, supporting local rock climbing development. Serves HRDEM (1m resolution) rasters, processed by edge-detection and machine learning models to extract relevant features.
Technologies used : Titiler, Cloud-Optimized GeoTiff, GDAL, Vite, Python.

PUBLICATIONS

S.J. Johnston, J. Murfitt, C. Duguay, & C. Albergel. (2025). “Addressing Cloud Contamination in Satellite Lake Ice Cover.” *In prep*.

S.J. Johnston, J. Murfitt, & C. Duguay. (2025). “Deep Learning for Lake Ice Cover Forecasting.” *Submitted to GMD*.

P.A. Costa Rocha, **S.J. Johnston**, et al. (2023). “DNN Modeling for CFD: Benchmarking the FNO.” *Appl. Sci.*, 13(5):3165. [doi:10.3390/app13053165](#)

N.B. Basu, J. Dony, K.J. Van Meter, **S.J. Johnston**, & A.T. Layton. (2023). “Random Forest in the Great Lakes: Stream Nutrients.” *Earth's Future*, 11(4). [doi:10.1029/2021EF002571](#)